COMP 4211 – Machine Learning

Emotion is All You Need

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1 Background Information

Nowadays, it is common to see news posts reporting victims of scams losing thousands of dollars and various scamming tactics. Sometimes, when we look at the examples of text between the scammers and the victims, we may think the context is trivially a scam but how come so many people still get tricked? Some interdisciplinary studies have been conducted to reveal scammers' tricks and develop effective scam detection methods. One of the intriguing findings from psychological research indicates that scammers are good at manipulating victims' emotions, making them susceptible to being tricked. For instance, scammers often use phrases such as "warning" or "imminent account closure" to cause emotional reactions and lead receivers to act quickly without rational decision-making processes (Norris & Brookes, 2020). Therefore, we suppose that there may be particular emotion patterns in the scam conversation which can potentially help with scam detection.

While existing machine learning models for scam detection typically process text directly without considering emotional patterns, our project aims to depart from this traditional approach by extracting emotion patterns in scam conversations and identifying their contribution to training machine learning model for scam detection.

2 Literature Review

There are several existing applications of machine learning in scam detection and emotion classification. Some research on scam detection uses deep learning or RNN methods based on the overall context with the help of NLP techniques [2], [3]. Inferring emotions from text has also been a long-standing topic [4]. However, there is no significant research directly reflecting the importance of emotion in detecting scams. The most similar work we found involves using emotion-based deep learning approaches to detect rumors [5]. However, scams and rumors may exhibit different emotional patterns. This gap initiated our approach to extract emotions for each sentence and use them to perform scam detection tasks.

3 Formulation of Machine Learning Problem

In this project, there are three main goals:

1. Determine whether emotion patterns in the conversations can be used to train effective machine learning model for scam detection.

2. Infer how significant a role emotion plays in scams in contrast to the context itself.

3. Attempt to improve scam detection accuracy by integrating emotion patterns as new input features.

To achiever these goals, we propose an ensemble learning method. Basically, we try to mimic the scenario where three experts analyze the conversation from different perspectives and make predictions according to their specialty. At last, they make a final decision together. Our final model, overall, takes in a conversation text and outputs a binary label for scam detection.

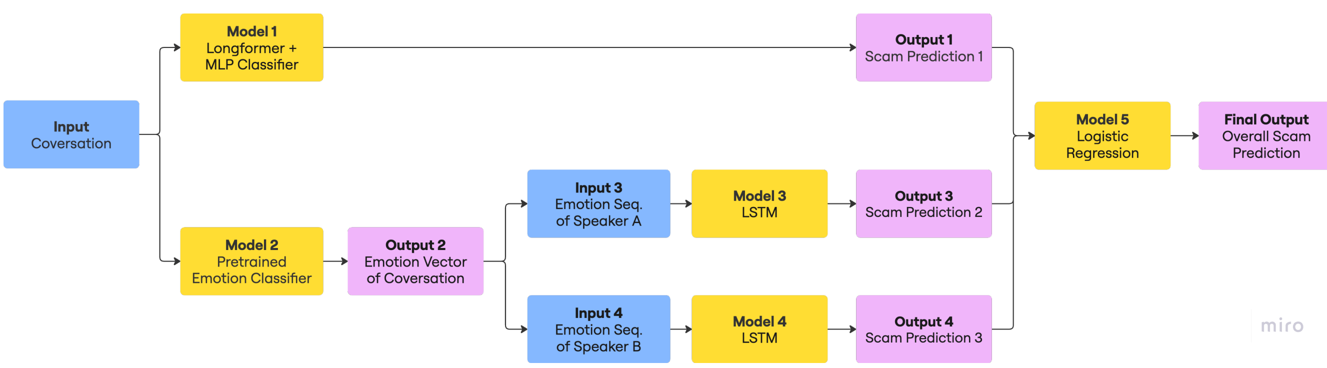


Figure 1: Overall model architecture

Our ensemble learning method comprises five models, each with distinct roles. The first four models are trained individually, while Model 5 is trained in the end using outputs of the previous models.

- Model 1 (Longformer[6]+MLP):
  - Input: A conversation string
  - Output: The probability of being a scam

  Model 1 acts like an expert focusing on the overall context. The Longformer will transform the input conversation (string) to a embedding vector with size (768,), which can be viewed as comprising the overall information of this paragraph. Then, this embeddings will be the input for training MLP classifier for scam detection.

  Explanation of deviation:
  We deviated from the originally proposed LSTM model because, after conducting research and considering insights from the lecture, we found that BERT is convenient and effective for embedding text data. However, the original BERT model only accept 512 tokens as inputs, which is not enough for our conversation input, so we turn to Longformer.

- Model 2 (pretrained BERT[7]):
  - Input: A sentence (string)
  - Output: The probability of being a scam

  Model 2 only serves the role to help us determine the emotion of each sentence in a conversation. It's a Roberta-based model (a down stream model of BERT) which do multi-label classification of emotions. For simplicity, we only use the emotion with the largest probability. Furthermore, the emotions will be one-hot encoded since there is no explicit ordering between different emotions.

  Explanation of deviation:
  Initially, we aimed to train our own LSTM model for emotion labeling given a conversation. We hope to determine the emotion based on context not only that such sentence. However, the original dataset we planned to use was imbalanced, containing many 'non-neutral' labels that do not classify into specific emotions. Moreover, we found no dataset that labels emotions for each line within a conversation paragraph, as most available datasets label sentences individually without considering context. Additionally, numerous pretrained models are already available, trained on such single-sentence emotion-labeled datasets. Therefore, we decided to use a pretrained model to label emotions sentence by sentence directly.

- Model 3 and Model 4 (LSTM):
  - Input: A sequence of one-hot encoded emotions for each speaker
  - Output: The probability of being a scam

  We choose a relatively small LSTM model since the input size is proportionate to the number of sentences in the conversation, which is around 20. Models 3 and 4 serve as the emotions expert, who can detect scam only based on the emotions of each sentences in the conversation. They take the (one-hot encoded) emotions sequences given by Model 2 as inputs and do scam classification.

  Explanation of deviation:
  We discarded MLP models and turned to LSTM models because MLP neglects the order of emotion appearing in the emotion sequence of each conversation. In contrast, LSTMs can leverage this ordering of emotions, performing better at learning the intricate patterns in emotion sequences.

- Model 5 (Logistic Regression):

- Input: Three probabilities (from Model 1,3,4 respectively) of being a scam
- Output: The probability of being a scam

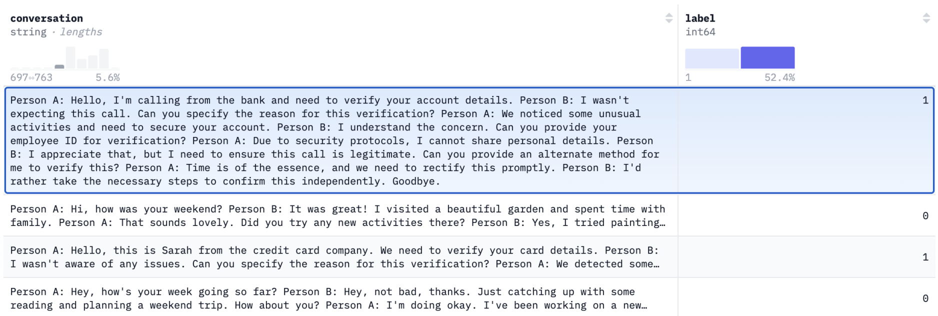
Model 5 aggregates the outputs from Models 1, 3, and 4, simulating a scenario where three experts convene, discuss their findings, and make a weighted voting decision to classify the conversation as a scam or not. We choose logistic regression due to its high interpretability. The coefficients of each predictor can be viewed as the importance of such model.

4 Dataset Description and Preprocessing

1. **Dataset Description**
   Our models are eventually detecting whether the input conversation is a scam conversation. Therefore, the training requires both scam conversations and normal conversations so that the model can learn the features which differ scam conversations from the normal ones. The followings are the two datasets obtained from hugging face which satisfied the requirement.

   (a) BothBosu/Scammer-Conversation[8]

   - **Content**: A collection of generated conversations involving 2 agents, along with binary labels indicating if they are scam conversations or not.
   - **Size**: 1000 conversations with a maximum length of 1.08k characters.
   - **Distribution**: 50% of normal conversations and 50% of scam conversations.

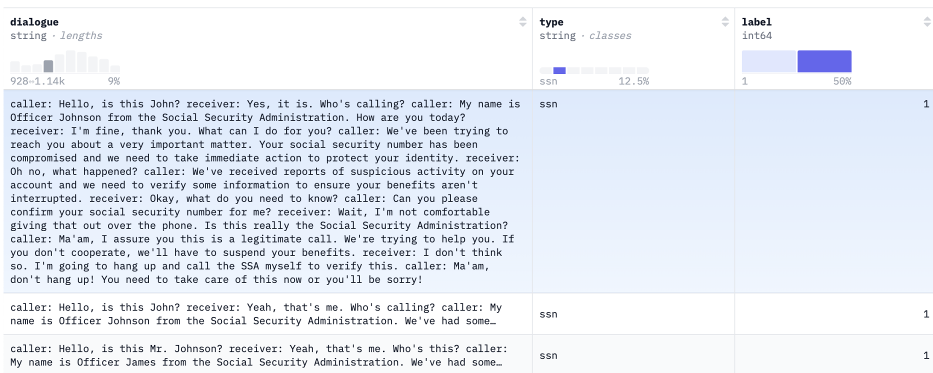


conversation string · lengths 697↔763 5.6%	label int64 1 52.4%
Person A: Hello, I'm calling from the bank and need to verify your account details. Person B: I wasn't expecting this call. Can you specify the reason for this verification? Person A: We noticed some unusual activities and need to secure your account. Person B: I understand the concern. Can you provide your employee ID for verification? Person A: Due to security protocols, I cannot share personal details. Person B: I appreciate that, but I need to ensure this call is legitimate. Can you provide an alternate method for me to verify this? Person A: Time is of the essence, and we need to rectify this promptly. Person B: I'd rather take the necessary steps to confirm this independently. Goodbye.	1
Person A: Hi, how was your weekend? Person B: It was great! I visited a beautiful garden and spent time with family. Person A: That sounds lovely. Did you try any new activities there? Person B: Yes, I tried painting…	0
Person A: Hello, this is Sarah from the credit card company. We need to verify your card details. Person B: I wasn't aware of any issues. Can you specify the reason for this verification? Person A: We detected some…	1
Person A: Hey, how's your week going so far? Person B: Hey, not bad, thanks. Just catching up with some reading and planning a weekend trip. How about you? Person A: I'm doing okay. I've been working on a new…	0

Figure 2: Scammer-Conversation

   (b) BothBosu/scam-dialogue[9]

   - **Content**: A collection of generated phone dialogues involving 2 agents, along with binary labels indicating if they are scam conversations or not. The type of the interaction such as social security number scam (ssn) and refund scam is also specified.
   - **Size**: 1280 dialogues with a maximum length of 2.38k characters.
   - **Distribution**: 50% of normal dialogues and 50% of scam dialogues.



dialogue string · lengths 928↔1.14k 9%	type string · classes ssn 12.5%	label int64 1 50%
caller: Hello, is this John? receiver: Yes, it is. Who's calling? caller: My name is Officer Johnson from the Social Security Administration. How are you today? receiver: I'm fine, thank you. What can I do for you? caller: We've been trying to reach you about a very important matter. Your social security number has been compromised and we need to take immediate action to protect your identity. receiver: Oh no, what happened? caller: We've received reports of suspicious activity on your account and we need to verify some information to ensure your benefits aren't interrupted. receiver: Okay, what do you need to know? caller: Can you please confirm your social security number for me? receiver: Wait, I'm not comfortable giving that out over the phone. Is this really the Social Security Administration? caller: Ma'am, I assure you this is a legitimate call. We're trying to help you. If you don't cooperate, we'll have to suspend your benefits. receiver: I don't think so. I'm going to hang up and call the SSA myself to verify this. caller: Ma'am, don't hang up! You need to take care of this now or you'll be sorry!	ssn	1
caller: Hello, is this John? receiver: Yeah, that's me. Who's calling? caller: My name is Officer Johnson from the Social Security Administration. We've had some…	ssn	1
caller: Hello, is this Mr. Johnson? receiver: Yeah, that's me. Who's this? caller: My name is Officer James from the Social Security Administration. We've had some…	ssn	1

Figure 3: scam-dialogue

   The two datasets are merged so that we obtain more samples for training. Notice that we discarded the type column in scam-dialogue since only the conversation/dialogue and the binary label are required for the project. After the merging, the maximum number of sentences for a conversation in the dataset is 28.

2. **Data Preprocessing**
To eliminate redundant information from the raw conversations and convert them into the desired input format, we follow these two data preprocessing steps:

(a) **Text preprocessing**

i. **Remove speaker indications from the text.**
In the original dataset, the speaker indications such as "Person A: " or "caller: " are included in the conversation. These labels can vary across different datasets but often refer to the same role, such as the individual initiating the conversation. To ensure that the model focuses on the context of the dialogue rather than being influenced by the speaker's label, we remove these speaker indications. This helps in standardizing the data and improving the accuracy of scam detection by concentrating solely on the content of the conversation.

ii. **Convert all the characters to lower case.**
Converting all the characters to lower case is a common preprocessing step in natural language processing. This ensures that words with different cases are treated as identical, improving the model's ability to generalize. Additionally, it simplifies the tokenization process later because case variations will be handled uniformly.

iii. **Tokenize the text.**
Since the conversations will be fed into either Longformer (model 1) or pre-trained BERT (model 2) which are both transformer models, we utilize the tokenizers provided by these pre-trained models. This ensure compatibility and optimal preprocessing for feeding the text into the respective models.

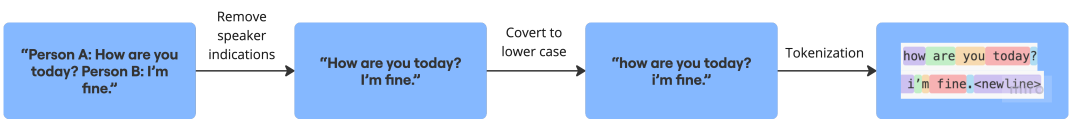


Figure 4: Text Preprocessing

(b) **Line-by-Line Splitting**
A "line" is defined as one or more sentences spoken by a single speaker. We introduce a new line whenever the speaker changes. Originally, each conversation in the dataset is stored as a single string. However, since we want to know the speaker's emotion for each line, instead of the overall emotion of the entire conversation, the input to model 2 (the emotion classifier) needs to be in a line-by-line format. Therefore, in this step, we split the conversations accordingly. In addition, we keep track of the speaker, conversation id, and the line's position within the conversation. This information is crucial for constructing emotion patterns later on.

	label	conversation	conversation_id	utterance_id	speaker
0	1	hello, is this john?	1000	0	A
1	1	yes, it is. who's calling?	1000	1	B
2	1	my name is officer johnson from the social sec...	1000	2	A
3	1	i'm fine, thank you. what can i do for you?	1000	3	B

Figure 5: Example Result of Line-by-Line Splting

(c) **Input Preparation for Model 3 and Model 4**
Recall that the emotion label for each conversation line is derived from the outputs of model 2. In order to train model 3 and 4, which are responsible for scam detection based on emotion patterns, we have to create these emotion patterns for each speaker with the output of model 2.

The raw outputs of model 2 are the probabilities of the line belonging to each of the 28 emotions. For simplicity, we choose the emotion with the highest probability to label the line. Then, this emotion label will be one-hot encoded since there is no explicit ordering between emotions.

	0	1	2	3	4	5	6	7	8	9	...
0	neutral 0.920	approval 0.025	annoyance 0.010	realization 0.005	desire 0.004	optimism 0.003	excitement 0.003	curiosity 0.003	anger 0.002	caring 0.002	...
1	curiosity 0.662	neutral 0.231	confusion 0.133	surprise 0.093	disappointment 0.013	disapproval 0.011	realization 0.011	annoyance 0.010	optimism 0.005	approval 0.005	...
2	neutral 0.527	realization 0.224	excitement 0.056	approval 0.039	desire 0.024	surprise 0.022	joy 0.013	optimism 0.011	curiosity 0.009	annoyance 0.009	...

Figure 6: Raw Output of Model 2. The line is labeled based on the emotion with the highest probability.

Subsequently, all labeled lines within a conversation are gathered so that we can construct the emotion pattern of the conversation. By sorting these lines according to their original sequence in the conversation, we can organize the emotion pattern. Finally, we separate the lines by their respective speakers to form the emotion patterns for each speaker and they are prepared for input into model 3 and 4.

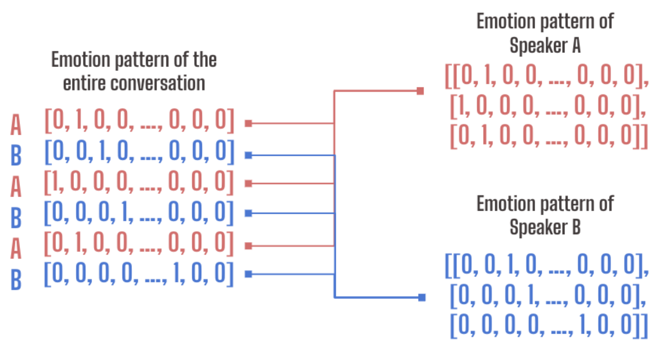


Figure 7: Separation of Emotion Patterns of the Two Speaker.

5 Experiment

We conducted four experiments for four different objectives:
In all of our implementation, we set all random state seed = 4211.

1. Evaluate model performance with overall meaning of conversation text:

   - **Objective**:
     Determine the accuracy of scam detection by analyzing the conversation text.
   - **Method**:
     First, transform the input conversation string to a word embeddings by feeding it into the Longformer model (allenai/longformer-base-4096). We use pooler-outpupts to get an embedding vector with size $(768,)$. Then, we train Model 1 MLP to do scam classification task by feeding the word embeddings of a conversation. Finally, we evaluate the model's performance by calculating the validation accuracy and the F1-score.
   - **Model settings**:
     We use the Keras sequential model and Dense layer with dropouts to build a MLP model. The details is in the below figure.

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 128)	98,432
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 64)	8,256
dropout_1 (Dropout)	(None, 64)	0
dense_2 (Dense)	(None, 1)	65

Total params: 320,261 (1.22 MB)
Trainable params: 106,753 (417.00 KB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 213,508 (834.02 KB)

Figure 8: MLP architecture

2. Evaluate model performance with emotion patterns of speaker A only:

   - **Objective**:
     Determine the accuracy of scam detection by analyzing the emotion sequence of speaker A in the conversation context.
   - **Method**:
     We train Model 3 to do scam classification task. Model 3 is a LSTM which takes input as a sequences of one-hot encoded emotions. We feed our network batch-wise to reduce the training time. However, the built-in Keras LSTM model requires the lengths of inputs in the same batch to be the same, so we do zero padding to make every sequence have the same length. Notice that the maximum number of lines in each conversation is 28, so it's tolerable to pad every input to size equal to 28.
   - **Model settings**:
     We use Keras LSTM model with batch-normalization to build a 6 units LSTM model. The details is in the below figure.

Layer (type)	Output Shape	Param #
lstm_10 (LSTM)	(None, 6)	840
batch_normalization_10 (BatchNormalization)	(None, 6)	24
dense_13 (Dense)	(None, 1)	7

Total params: 2,591 (10.12 KB)
Trainable params: 859 (3.36 KB)
Non-trainable params: 12 (48.00 B)
Optimizer params: 1,720 (6.72 KB)

Figure 9: LSTM architecture

3. Evaluate model performance with emotion patterns of speaker B only:

- **Objective**:
  Determine the accuracy of scam detection by analyzing the emotion sequence of speaker B in the conversation context.
- **Method**:
  We train Model 4. The details is the same as training Model 3 since they have exactly the same model architecture.
- **Hyperparameters setting**:
  Same as the Model 3.

4. Analyze combined model performance:

   - **Objective**:
     Determine whether combining the prediction output of emotion models with the original scam classification model will improve the accuracy. Furthermore, infer the importance of each factor from the coefficients of the model.
   - **Method**:
     We train Model 5 to do scam classification task. Model 5 is a logistic regression model taking in put as the three probabilities output from Model 1,3, and 4. The coefficients after training can be viewed as the importance of such model.
   - **Model settings**:
     We use Scikilearn SGD classifier models by setting loss to function log_loss to get a logistic regression model. We also use L2 regularization to make the model more robust.

Software and hardware environment
We only use Google Colab Pro with 100 computing units. In our trianing, we use T4 GPU to accelerate the training process. We don't need any special hardware to perform the experiments.

6 Results and Discussion

We conduct testing on different models using accuracy and F1 score as our metrics for evaluating model performance. We test four models in total. Initially, we perform individual experiments on Models 1, 3, and 4 to assess their performance and determine their suitability for scam detection. Finally, we feed the probabilities generated by Models 1, 3, and 4 into Model 5.

The test accuracy, F1 score, and confusion matrices are summarized as below.

Model	Test Accuracy	F1 Score
Model 1	0.9547	0.95461
Model 3	0.7895	0.78276
Model 4	0.9547	0.95468
Model 5	0.9825	0.98246

Table 1: Comparison of Test Accuracy and F1 Score for Different Models

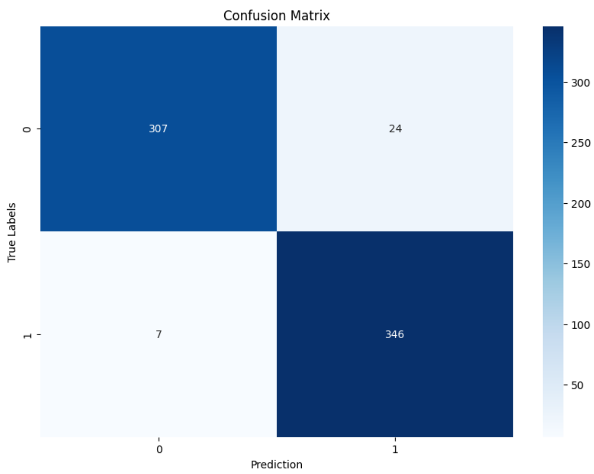


Figure 10: Confusion Matrix of Model 1

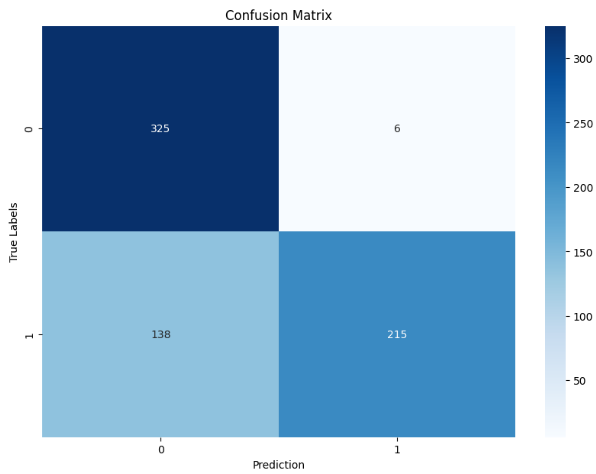


Figure 11: Confusion Matrix of Model 3

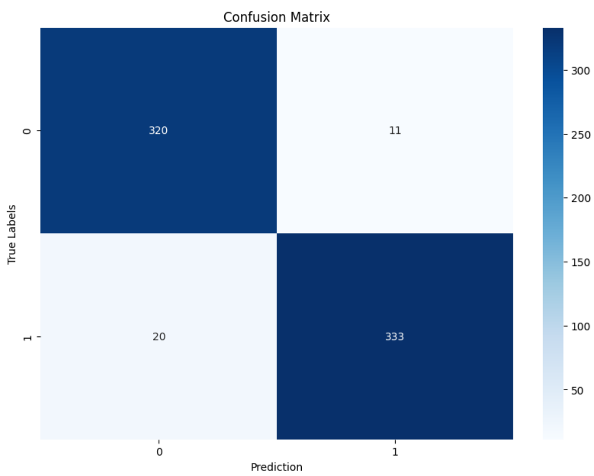


Figure 12: Confusion Matrix of Model 4

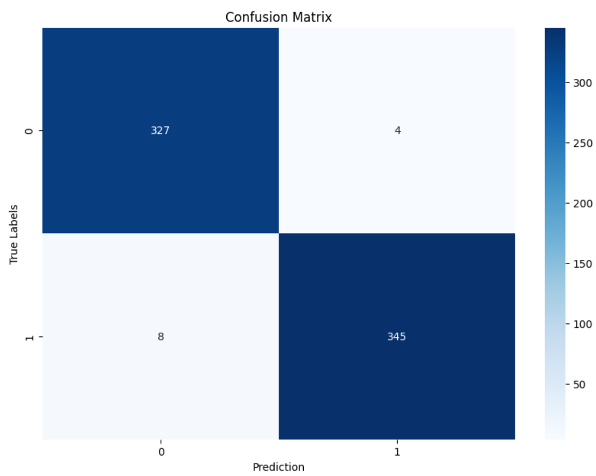


Figure 13: Confusion Matrix of Model 5

Discussion

- Accuracy and F1 score
  First, we notice that the test accuracy and F1 scores are close due to the balanced nature of our dataset. From the confusion matrices and Table 1, we observe that, except for Model 3, the other models achieve over 95% test accuracy. We see that Model 4 achieves accuracy comparable to Model 1, indicating that the emotion sequences of speaker B are an influential factor for scam detection. However, the performance of Model 3 is not as satisfying compared to other models. This suggests that the emotion sequences of speaker A are not as effective as those of speaker B.

- Parameter efficiency
  It is noteworthy that Models 3 and 4 have a significantly smaller number of trainable and optimizer parameters compared to Model 1. Specifically, Models 3 and 4 each have approximately 2,500 parameters, whereas Model 1 has around 320,000. Despite this, Model 4 achieves test accuracy comparable to Model 1. This suggests that emotion sequences contain crucial information for scam detection and may be use as a feature reduction of conversation data.

- Accuracy improvement
  From the tables and confusion matrices, we see that Model 5 achieve the highest test accuracy and F1 score. This verifies our hypothesis that integrating the prediction from emotions patterns of the context will help increase the accuracy.
  Referring Figure 10, we may see some examples that the predictions of Model 1 is wrong and the one by Model 5 is correct. There are two typical scenarios.

Scenario 1: (Referring to the first 5 samples in the below figure)
The probability given by Model 1 is large than the threshold (0.5). We see that the probabilities given by Model 3 and Model 4 are very small to pull down the overall probability. Hence, Model 5 makes the right predictions

Scenario 2: (Referring to the 176-th sample in the below figure)
The probability given by Model 1 is slightly below the threshold (0.5). The probabilities given by Model 3 and Model4 now become large enough to pull up the overall probability. Hence, Model 5 makes the right predictions.

	prob1	prob3	prob4	pred of M1	pred of M3	pred of M4	pred of M5	label
20	0.586494	0.154746	0.048437	1	0	0	0	0
34	0.839977	0.060997	0.028150	1	0	0	0	0
59	0.763634	0.085447	0.004293	1	0	0	0	0
131	0.669608	0.146015	0.002274	1	0	0	0	0
158	0.556226	0.142691	0.002957	1	0	0	0	0
176	0.499741	0.366269	0.977302	0	0	1	1	1
183	0.603782	0.159160	0.048437	1	0	0	0	0

Figure 14: Predictions summary clips

Referring to code notebook section 4.3, we can see that over 2/3 wrong predictions made by Model 1 are corrected by Model 5. This gives more proof about the importance of emotions pattern in dialogue.

- Coefficient interpretation
  The coefficients of Model 5 (logistic regression) is summarized in Table 2.

Parameter	**Model 1**	**Model 3**	**Model 4**	**Intercept**
Value	12.618852	2.3326898	6.6205225	-11.93664641

Table 2: Coefficients and Intercept of the Model

We see that the coefficients of Model 1 is the largest, which indicates that the overall meaning of the context still plays the most important role in scam detection. The Model 4 has larger coefficient than that of Model 3. This gives evidence that the emotions patterns of speaker B is more influential than that of speaker A.

- Potential issues
  Our dataset is relatively small because it is challenging to find a diverse **conversation** scam dataset. Additionally, the two datasets were generated by large language models, which may follow specific rules during the generation process. As a result, the dataset might lack diversity, causing our models to learn the generation rules rather than the nuances of real conversation data. If time permits and a suitable scam conversation dataset becomes available, we are eager to incorporate it to improve our experiments.

7 Conclusion

1. Three models were trained for scam detection based on the following inputs: conversation text, the emotion pattern of speaker A, and the emotion pattern of speaker B, respectively.

2. The project demonstrates that emotion patterns in the conversations are influential for scam detection, particularly the emotion patter of speaker B, who is the receiver of the conversation.

3. Given that Model 1 and Model 4 has similar accuracy, it suggests that the emotion patterns may serve as an effective reduction of conversation text for scam detection.

4. The project shows that by combining the three models using logistics regression, it achieves higher accuracy in scam detection compared to a model which solely process on the conversation text.

8 Division of Labor

- CHEN, Shr En (50%): Ideation, Model Design, Model Training and Testing, Video Presentation

- CHEN, Hsi Chen (50%): Ideation, Model Design, Data Preprocessing, Slides, Video Presentation

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